Total Questions: 46 (to be updated)

1. Explain Logistic Regression and how it is different from Linear Regression

Logistic Regression is a type of statistical method used to predict a binary outcome, which means it predicts whether something is "yes" or "no", "true" or "false", etc. It does this by using a special function called the sigmoid function, which squeezes the output of a linear equation between 0 and 1.

Linear Regression, on the other hand, is used for predicting a continuous outcome, like predicting someone's salary based on their years of experience. It tries to fit a straight line through the data.

The key difference is that Logistic Regression is used for classification (like "yes" or "no"), while Linear Regression is used for predicting numerical values.

2. What is Log Loss?

Log Loss, also known as logistic loss or cross-entropy loss, is a measure of how good your classification model is at predicting the correct outcome. It penalizes predictions that are confident but wrong more than predictions that are uncertain but correct. A lower log loss means the model is performing better.

3. What is the Optimization Algorithm in Logistic Regression?

The optimization algorithm commonly used in Logistic Regression is called Gradient Descent. Gradient Descent is a method for finding the best coefficients (weights) for your model by minimizing the error (or loss). It does this by iteratively adjusting the weights to reduce the difference between the predicted and actual values.

4. What are the steps of preprocessing? Why is scaling important? Why is KNN sensitive to scaling?

Preprocessing steps typically include:

1. **Handling Missing Values**: Filling in or removing missing data.
2. Encoding Categorical Variables: Converting text labels into numbers.
3. Scaling: Standardizing or normalizing the data so that all features have similar scales.
4. Splitting Data: Dividing the data into training and testing sets.

Scaling is important because many algorithms, including KNN (K-Nearest Neighbors), calculate distances between data points. If one feature has a much larger scale than others, it can dominate the distance calculation, leading to inaccurate predictions.

KNN is sensitive to scaling because it relies on distance to find the nearest neighbors. If features are not scaled, those with larger values will disproportionately affect the distance, making the algorithm less accurate.

5. Explain overfitting and underfitting with an example

Overfitting happens when a model learns not only the underlying pattern but also the noise in the data. This means the model works very well on the training data but fails to generalize to new data. Imagine you are studying for an exam by memorizing every question from past papers. You might do well on those exact questions but struggle with new ones.

Underfitting occurs when the model is too simple and doesn't capture the underlying pattern of the data. It's like trying to learn a complex topic by only reading a summary (like I am doing for this Viva). You won't have enough knowledge to answer even the basic questions.

6. Explain step by step, the working of K-means Clustering

K-means Clustering is a method used to group similar data points together into clusters. Here's how it works:

1. Choose the number of clusters (K): You decide how many groups you want.
2. Place K centroids: Initially, pick K random points as the center of each cluster.
3. Assign points to the nearest centroid: Each data point is assigned to the cluster with the nearest centroid.
4. Update centroids: Calculate the new centroids by finding the mean of all the points in each cluster.
5. Repeat: Steps 3 and 4 are repeated until the centroids don't change much, meaning the clusters are stable.

7. What is the Elbow Method?

The Elbow Method is a technique used to find the optimal number of clusters in K-means clustering. You plot the number of clusters (K) on the x-axis and the within-cluster sum of squares (a measure of how compact the clusters are) on the y-axis. The "elbow" point, where the curve starts to flatten, is often considered the best choice for K because it represents a good balance between having too few and too many clusters.

8. What is dimensionality reduction, why is it important?

Dimensionality reduction is the process of reducing the number of input variables (features) in your dataset. It's important because high-dimensional data can be very complex, hard to visualize, and computationally expensive to process. Reducing dimensions can help make the data easier to work with and can improve the performance of machine learning models.

9. How does PCA work?

Principal Component Analysis (PCA) is a technique used for dimensionality reduction. It works by:

1. Standardizing the data: Making sure each feature has a similar scale.
2. Finding the principal components: These are new features that are combinations of the original ones, but they capture the most variance (or information) in the data.
3. Transforming the data: The original data is transformed into this new set of features, which are fewer in number but still contain most of the important information.
4. Choosing the top components: You can then keep only the top components, which reduces the dimensions while retaining as much information as possible.

This makes the data simpler and often speeds up machine learning models.

10. Free Question

Drink water

11. What is preprocessing?

Preprocessing is the process of transforming raw data into a clean and suitable format for analysis or machine learning. This includes tasks like handling missing data, encoding categorical variables, scaling features, and splitting data into training and testing sets. Preprocessing helps to improve the quality of the data, which in turn can improve the performance of models.

12. What is EDA?

EDA stands for Exploratory Data Analysis. It is the process of analyzing datasets to summarize their main characteristics, often using visual methods. EDA helps you understand the data better, identify patterns, spot anomalies, and check assumptions. It usually involves plotting graphs, calculating basic statistics, and looking for correlations.

13. What EDA have you used?

Common EDA techniques include:
- Descriptive statistics: Mean, median, mode, standard deviation.
- Data visualization: Histograms, scatter plots, box plots, bar charts, heatmaps.
- Correlation analysis: Checking the correlation between different variables.
- Outlier detection: Identifying unusual data points that don't fit the general pattern.

14. What is imputation?

Imputation is the process of filling in missing values in a dataset. Instead of removing rows with missing data, imputation allows you to replace missing values with estimated ones, such as the

mean, median, or mode of the column, or using more advanced methods like predictive modeling.

15. What is scaling?

Scaling is the process of adjusting the range of features in your data so they have a similar scale. This is important for algorithms that rely on distance calculations (like KNN or SVM). Common scaling techniques include:
- Normalization: Rescaling data to a range of [0, 1].
- Standardization: Adjusting data to have a mean of 0 and a standard deviation of 1.

16. What is dimensionality reduction?

Dimensionality reduction is the process of reducing the number of input features in a dataset. It helps in simplifying models, reducing computational cost, and avoiding the curse of dimensionality, which can occur when dealing with high-dimensional data.

17. What is PCA?

Principal Component Analysis (PCA) is a technique for dimensionality reduction. It transforms the original features into a smaller set of new features (called principal components) that still capture most of the variability in the data. PCA helps in reducing complexity while retaining the most important information in the dataset.

18. What is K-means? Does it change the size of the dataset? How exactly does it work?

K-means is a clustering algorithm used to group similar data points into clusters. It doesn't change the size of the dataset; instead, it assigns each data point to one of the clusters.

How it works:
1. Select K: Decide the number of clusters.
2. Initialize centroids: Choose K random points as the initial centroids.
3. Assign clusters: Assign each data point to the nearest centroid.
4. Update centroids: Recalculate the centroids based on the points in each cluster.
5. Repeat: Steps 3 and 4 until the centroids stabilize.

19. What is KNN?

K-Nearest Neighbors (KNN) is a simple, non-parametric classification (or regression) algorithm. It works by finding the K nearest data points to a new observation and predicting its class (or value) based on the majority class (or average value) among the neighbors.

20. What are its detailed steps?

1. Choose K: Decide the number of neighbors to consider.
2. Calculate distances: Measure the distance between the new data point and all other points in the dataset.
3. Find nearest neighbors: Identify the K points closest to the new data point.
4. Make prediction: For classification, assign the most common class among the neighbors; for regression, take the average value.

21. What is a loss function?

A loss function measures how well a machine learning model's predictions match the actual data. It quantifies the difference between the predicted output and the true output. The goal during training is to minimize the loss function.

22. What is log loss?

Log Loss (or logistic loss) is a specific loss function used for binary classification. It penalizes predictions that are confident but wrong more than predictions that are uncertain but correct. A lower log loss indicates a better-performing model.

23. What is linear regression? How is it different from logistic regression?

Linear Regression is a method for predicting a continuous outcome (e.g., predicting a price) based on input features. It tries to fit a straight line to the data.

Logistic Regression, on the other hand, is used for predicting a binary outcome (e.g., yes/no). It uses the logistic function to ensure the predictions are probabilities between 0 and 1.

The main difference is that linear regression is for continuous outcomes, while logistic regression is for binary classification.

24. What is overfitting and underfitting? Explain with an example.

- Overfitting happens when a model is too complex and captures not only the underlying pattern in the data but also the noise. It performs well on the training data but poorly on new data. For example, if you memorize answers to practice questions but can't answer different questions in the exam, you're overfitting.

- Underfitting happens when a model is too simple and fails to capture the underlying pattern in the data. It performs poorly on both training and new data. For example, if you only read the summary of a textbook, you may not know enough to answer questions in detail.

25. What is a decision tree?

A Decision Tree is a model that makes predictions by splitting data into subsets based on certain conditions. It forms a tree-like structure where each internal node represents a decision based on a feature, each branch represents the outcome of that decision, and each leaf node represents a final prediction or outcome.

26. What is bagging and boosting?

- Bagging (Bootstrap Aggregating) is an ensemble technique where multiple models (usually of the same type) are trained on different subsets of the data, and their predictions are averaged. This reduces variance and helps prevent overfitting.

- Boosting is another ensemble technique where models are trained sequentially, each trying to correct the errors of the previous one. The final prediction is a weighted sum of all models, which helps improve accuracy by reducing bias and variance.

27. What can you conclude from a confusion matrix/classification report?

A Confusion Matrix is a table that shows the true positive, true negative, false positive, and false negative outcomes of a classification model. It helps in understanding how well the model is performing in terms of accuracy, precision, recall, and F1-score.

A Classification Report provides these metrics in more detail, allowing you to assess the model's performance on each class separately.

28. Explain the ROC curve. What can you conclude from it?

The ROC (Receiver Operating Characteristic) Curve is a plot that shows the true positive rate (sensitivity) against the false positive rate (1-specificity) at different threshold settings. The area under the curve (AUC) represents the model's ability to distinguish between classes. A higher AUC indicates a better-performing model. The ROC curve helps in selecting the optimal threshold for classification by balancing sensitivity and specificity.

29. What is Logistic Regression?

Logistic Regression is a classification algorithm used to predict the probability of a binary outcome (e.g., yes/no, true/false). It uses a logistic function (or sigmoid function) to convert the linear combination of input features into a probability value between 0 and 1.

30. What is KNN?

KNN, or K-Nearest Neighbors, is a simple, non-parametric algorithm used for classification and regression. It works by finding the K nearest data points to a new observation and predicting its class based on the majority class among those neighbors.

31. Explain K-means Algorithm

K-means is a clustering algorithm that groups data into K clusters based on similarity. It works as follows:

1. Choose the number of clusters (K).
2. Randomly select K initial centroids.
3. Assign each data point to the nearest centroid.
4. Recalculate the centroids based on the points assigned to each cluster.
5. Repeat the assignment and centroid update steps until the centroids no longer change significantly.

32. What is Preprocessing?

Preprocessing refers to the steps taken to clean and prepare raw data for analysis or machine learning. This might include handling missing data, encoding categorical variables, scaling features, and splitting data into training and testing sets.

33. Why do you preprocess?
Preprocessing is crucial because raw data often contains noise, missing values, and inconsistencies. Proper preprocessing ensures that the data is clean, consistent, and ready for analysis or modeling, leading to better model performance.

34. Give some methods of preprocessing and explain
- Handling Missing Values: You can either remove rows/columns with missing data or fill (impute) the missing values using techniques like mean/mode/median imputation.
- Encoding Categorical Variables: Converting text labels into numerical values using techniques like one-hot encoding or label encoding.
- Scaling: Normalizing or standardizing features so that they are on a similar scale, which is important for algorithms that rely on distance calculations.
- Splitting Data: Dividing the dataset into training and testing sets to evaluate model performance.

34. Given a classification problem, how will you classify products?
To classify products, I would first preprocess the data, perform EDA, and then choose a suitable classification algorithm. Depending on the complexity and size of the dataset, I might use algorithms like Logistic Regression, Decision Trees, Random Forest, or even deep learning models. The choice depends on the nature of the data and the problem requirements.

35. What do you mean by loss function?
A loss function measures how well a machine learning model's predictions match the actual outcomes. It quantifies the error between the predicted and true values, guiding the model's learning process. The goal is to minimize the loss function during training.

36. Explain underfitting and overfitting
- Underfitting occurs when a model is too simple and fails to capture the underlying pattern in the data, leading to poor performance on both training and testing data.
- Overfitting happens when a model is too complex and captures not only the pattern but also the noise in the training data, resulting in excellent performance on the training data but poor generalization to new, unseen data.

37. Show the notebook and what algorithms did you use
In a practical setting, I would walk through a Jupyter notebook, showing the preprocessing steps, EDA, and the different algorithms tested. I would discuss why certain algorithms performed better, focusing on metrics like accuracy, precision, recall, and AUC.

38. Why did Logistic Regression perform well, and why didn’t others?
Logistic Regression might perform well because it is a simple model that works effectively with linearly separable data. It’s also less prone to overfitting compared to more complex models. Other algorithms might not perform as well if the data is not complex enough to require their added capacity or if they overfit the data due to high variance.

39. What is PCA?
Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms the original features into a smaller set of new features (principal components) that capture most of the variability in the data.

40. What is dimensionality reduction?
Dimensionality reduction is the process of reducing the number of input features in a dataset while retaining as much information as possible. This simplifies the model, reduces computational cost, and helps prevent overfitting.

41. Why is scaling important?
Scaling is important because many machine learning algorithms, such as KNN, SVM, and Gradient Descent-based methods, are sensitive to the scale of the features. If features are on different scales, it can lead to poor performance or convergence issues.

42. What is feature scaling?
Feature scaling is the process of normalizing or standardizing the range of independent variables (features) so that they contribute equally to the model. Common methods include min-max scaling and standardization (z-score normalization).

43. What are hyperparameters?
Hyperparameters are the settings or configurations that are set before training a machine learning model, such as learning rate, number of trees in a Random Forest, or the value of K in KNN. They control the learning process and need to be tuned for optimal performance.

44. Methods to do hyperparameter tuning
- Grid Search: Exhaustively searching through a specified set of hyperparameters.
- Random Search: Randomly sampling hyperparameters from a distribution.
- Bayesian Optimization: Using probabilistic models to guide the search for the best hyperparameters.

45. Explain the difference between the methods mentioned
- Grid Search is thorough but computationally expensive as it checks all possible combinations.
- Random Search is more efficient as it explores a broader range of values by sampling randomly, though it may miss the best combination.
- Bayesian Optimization is smarter and often faster because it uses past evaluations to choose the next set of hyperparameters to try, focusing on promising areas of the search space.

46. Explain ROC curve. What can you conclude from it?
The ROC (Receiver Operating Characteristic) Curve is a graph that shows the trade-off between the true positive rate (sensitivity) and the false positive rate (1-specificity) at different threshold levels. The Area Under the Curve (AUC) is a measure of the model's ability to distinguish between classes. A higher AUC indicates a better-performing model. The curve helps to choose the optimal threshold that balances sensitivity and specificity for your specific application.